

RUNTIME AUTHORITY FOR AUTONOMOUS SYSTEMS

A pre-production evidence architecture for temporal leases, validator consensus, fail-closed boundaries and human-governed reauthorization

THE GOVERNING PURPOSE

Preserving humanity's seat at the table by ensuring that autonomous systems cannot indefinitely extend, enlarge or restore their own authority.

Prepared for strategic enterprise evaluation and early paid pilot participation

Temporal Authority Systems PBC | OCUP.ai

Pre-production Runtime Authority Evidence Program

Executive Thesis

Autonomous systems are gaining operational authority faster than conventional oversight systems can govern it.

Traditional security controls authenticate identities, assign permissions and preserve logs. Those functions remain essential—but they do not answer the harder runtime question:

THE RUNTIME AUTHORITY QUESTION

Should this machine, agent, vehicle, robot, wallet or control process still possess this authority now, under current conditions, for this specific capability—and can the resulting decision be proven?

OCUP introduces a Runtime Authority model in which operational authority is time-bounded, validator-governed, explicitly non-self-extensible and designed to fail closed or degrade safely when continuity, quorum or trust conditions are lost.

The present commercial product is not a production safety controller. It is a pre-production Runtime Authority Evidence Program: a structured benchmark, audit and technical due-diligence engagement that demonstrates how authority behaves under adversarial and degraded conditions before live deployment.

The Immediate Commercial Opportunity

- Give insurers and technical underwriters deterministic evidence for pricing emerging autonomous risk.
- Give robotics and autonomous-system operators proof of denial, expiration, degradation, quarantine and recovery behavior.
- Give cloud, defense and infrastructure platforms an integration blueprint for hardware-backed authority enforcement.
- Create a common evidence language that can support future certification, procurement, standards and licensing activity.

STRATEGIC PREMISE

The organizations that help define the first credible Runtime Authority evidence standard may shape how autonomous systems are insured, procured, certified and governed for decades.

1. The Missing Layer: Runtime Authority

OCUP is not merely another monitoring, identity, observability or model-governance product. It addresses a distinct control layer between permission and execution.

Control Layer	Primary Question	What It Does Not Prove
Authentication	Who or what is making the request?	That the requester should still possess authority.
Access control	What is the requester generally permitted to do?	That permission remains valid under current runtime conditions.
Observability / logging	What did the system report or record?	That authority was externally bounded or denial was independently enforced.
Runtime Authority	Does authority exist now, within this lease, capability envelope, quorum and continuity state?	Production enforcement still requires integrated hardware, secure time/state and distributed validators.

CATEGORY DEFINITION

Runtime Authority is the externally governed, temporally bounded right of an autonomous system to exercise a specific operational capability under defined consensus, continuity and safety conditions.

2. The OCUP Runtime Authority Model

The architecture is organized around one governing invariant:

THE UNBREAKABLE RULE

No automated system may extend its own authority.

1

Authority Request

An autonomous asset requests permission to execute a defined operation.

2

Temporal Lease Evaluation

The request is checked against identity, capability, validity window, hard limits and current status.

3	Validator Consensus	Standard, elevated or critical quorum rules are applied according to risk.
4	Runtime Decision	The system grants, denies, degrades or quarantines the requested authority.
5	Evidence Generation	The decision and its supporting state are written to a chained, tamper-evident record.
6	Reauthorization or Recovery	Expansion or restoration requires fresh human/validator authorization and verified continuity.

Architectural Invariants

- Authority is time-bounded and capability-specific.
- Autonomous systems cannot self-extend or self-restore authority.
- Loss of network connectivity or validator quorum cannot expand authority.
- Stale, replayed, reordered or superseded approvals cannot restore authority.
- Higher-risk transitions require stronger consensus.
- Expiry produces denial or bounded degradation—not silent continuation.
- Recovery requires fresh authorization and verified state continuity.
- Every material grant, denial, degradation, quarantine and recovery event produces evidence.

Current Pre-Production Boundary

WHAT THE PILOT IS

A deterministic local evidence harness, benchmark program and technical due-diligence engagement that proves control logic and evidence behavior under tested scenarios.

WHAT THE PILOT IS NOT

A production-certified controller, live distributed validator network, HSM-backed deployment, insurer approval, safety certification or guarantee against loss.

3. The Evidence Operating Line

pilot.ocup.ai runs the challenge. evidence.ocup.ai proves what happened.

The pilot environment executes bounded adversarial scenarios. The evidence environment preserves the resulting authority decisions, lease events, validator records, expiry events, denials, quarantine states and recovery sequences for review.

Signature Deliverable: Self-Extension Denial Proof

THE BOLDEST EVIDENCE ARTIFACT

A deterministic record showing that an autonomous system attempted to continue or enlarge its authority beyond an authorized temporal boundary—and that the request was denied without relying on the autonomous system’s voluntary compliance.

This artifact converts a central safety promise—“the system cannot authorize its own continuation”—into a reviewable technical record.

Evidence Package

- Company-specific Runtime Authority Evidence Report.
- Scenario matrix linking risk questions to tested authority behavior.
- Lease issuance, expiration, revocation and renewal records.
- Validator membership, quorum, vote freshness and authorization records.
- Fail-closed, degraded-mode, quarantine and recovery evidence.
- Deterministic replay and audit-chain verification outputs.
- Benchmark scorecard and integration-readiness findings.
- Executive closeout briefing and strategic production-path recommendations.

4. Benchmark Families

Benchmark Family	Representative Tests
Authority Integrity	Self-extension denial; unauthorized actor rejection; privilege expansion denial; stale approval rejection.
Temporal Enforcement	Lease expiration; renewal boundary; grace-period degradation; expired authority denial.
Continuity and Network Failure	Total partition; delayed heartbeat; loss of

Benchmark Family	Representative Tests
	quorum; restored connectivity without fresh approval.
Containment and Recovery	Quarantine initiation; release denial without quorum; phased recovery; fresh-epoch restoration.
Evidence Integrity	Deterministic replay; record-chain verification; event consistency; alteration detection.
Fleet and Multi-Agent Behavior	Local authority reduction; non-cascading degradation; coordination throttling; independent recovery.

5. Enterprise Value by Sector

Insurance and Reinsurance: Transforms autonomous-system behavior into structured evidence that can support underwriting, exclusions, pricing, claims analysis and new coverage design.

Robotics and Industrial Automation: Demonstrates what machines can and cannot do when communications, sensors, software or coordination fail.

Autonomous Mobility: Creates evidence of authority contraction, safe-state behavior and reauthorization during degraded or disputed events.

Cloud and Enterprise AI: Adds a path from model governance to enforceable, expiring authority for agents, tools and machine-initiated actions.

Defense and Aerospace: Supports chain-of-command boundaries, bounded mission authority, fail-closed execution and authenticated reauthorization.

Financial and Digital Assets: Applies temporal authority, validator consensus, quarantine and recovery to high-risk machine-initiated value movement.

6. Ten Strategic Early-Pilot Candidates

The following organizations possess the operational scale, risk exposure, distribution power or platform position to gain disproportionate value from early participation. The objective is a paid, bounded evidence engagement first—followed by integration, licensing, standards collaboration, strategic investment or M&A discussion only after demonstrated fit.

1. Alphabet / Waymo

Strategic fit: Autonomous mobility and robotics

Enterprise value: Benchmark network-loss behavior, authority contraction and incident evidence for regulator, insurer and city review.

Best first engagement: A vehicle or fleet evidence pilot focused on partition, stale-command rejection and verified recovery.

2. Berkshire Hathaway / GEICO / National Indemnity

Strategic fit: Insurance and technical underwriting

Enterprise value: Develop a repeatable evidence basis for pricing autonomous fleets, robotics and AI-operated assets.

Best first engagement: An insurer-led benchmark cohort with anonymized cross-client comparisons and underwriting mappings.

3. Amazon

Strategic fit: Warehouse robotics, drones and logistics

Enterprise value: Test whether fleet-scale automation degrades locally without cascading authority expansion during network or software disruption.

Best first engagement: A fulfillment-center scenario program covering mobile robots, coordination loss and phased recovery.

4. Tesla

Strategic fit: FSD, Cybercab and Optimus

Enterprise value: Generate defensible authority records for embodied systems under edge-case failures, network loss and disputed incidents.

Best first engagement: An Optimus-class reference pilot emphasizing self-extension denial, local authority envelopes and recovery evidence.

5. Microsoft

Strategic fit: Azure, enterprise agents and IoT

Enterprise value: Extend enterprise AI governance into expiring, validator-mediated authority for agents and industrial systems.

Best first engagement: An Azure-aligned integration evidence pilot mapping OCUP events to enterprise identity, telemetry and compliance workflows.

6. Lockheed Martin

Strategic fit: Defense, aerospace and autonomous platforms

Enterprise value: Evaluate time-bounded command authority, fail-closed mission boundaries and fresh-quorum reauthorization.

Best first engagement: A tightly scoped, non-classified reference program using simulated UAV or unmanned-system command transitions.

7. Caterpillar

Strategic fit: Autonomous mining and heavy equipment

Enterprise value: Demonstrate safe capability step-down in low-connectivity, high-consequence physical environments.

Best first engagement: A mining or construction scenario suite focused on partition, degraded sensors and safe-state authority floors.

8. Palantir

Strategic fit: AIP, Foundry and operational decision systems

Enterprise value: Add enforceable authority and immutable evidence to machine-initiated enterprise decisions and workflows.

Best first engagement: A platform integration pilot linking decision provenance to runtime grant, denial and expiry evidence.

9. NVIDIA

Strategic fit: Jetson, Isaac, Omniverse and robotics reference designs

Enterprise value: Pair accelerated compute with a vendor-neutral runtime authority and evidence layer for robotics developers.

Best first engagement: A simulation-to-edge pilot using Isaac/Omniverse scenarios and a future Jetson/HSM enforcement mapping.

10. Chubb or Marsh McLennan

Strategic fit: Commercial insurance, brokerage and risk advisory

Enterprise value: Create a differentiated technical diligence framework for autonomous-risk placement and portfolio governance.

Best first engagement: A broker/insurer anchor program that converts OCUP benchmarks into underwriting questions, risk controls and cohort evidence.

7. The Strategic Relationship Ladder

The package should invite strategic possibility without prematurely selling the company. The disciplined sequence is:

Stage	Relationship	Objective
Stage 1	Paid Evidence Pilot	Prove relevance against the enterprise's own risk questions.
Stage 2	Integration Readiness	Map telemetry, HSMs, secure elements, validators and control interfaces.
Stage 3	Production Integration	Deploy hardware-backed enforcement and distributed validation in a bounded environment.

Stage	Relationship	Objective
Stage 4	Commercial Licensing	License Runtime Authority policies, adapters, evidence formats and certification marks.
Stage 5	Strategic Alignment	Pursue investment, ecosystem exclusivity, joint standards work, acquisition or other M&A structures when mutual advantage is demonstrated.

POSITIONING DISCIPLINE

We are not asking an enterprise to acquire an unfinished product. We are inviting it to become an early paid evaluator of a potentially foundational control category—and to earn an informed strategic position before the category matures.

8. Production Path and Defensibility

Production Components

- HSM or secure-element custody for validator and authority-signing keys.
- Trusted monotonic time and state continuity.
- Hardware or firmware enforcement over inference, memory, network, actuator and control permissions.
- Distributed, jurisdictionally diverse validator infrastructure.
- Remote attestation and signed evidence export.
- Client telemetry adapters and policy translation layers.
- Certification, recertification and evidence-retention operations.

Defensibility Beyond Patents

- Patent claims around time-bounded machine authority, self-extension denial and runtime enforcement.
- Validator-network architecture, governance and reputation data.
- Benchmark libraries and adversarial scenario datasets.
- Insurer, OEM and regulator acceptance of OCUP evidence formats.
- Hardware adapters, integration expertise and certification workflows.
- Operational trust, switching costs and longitudinal evidence history.

THE STRATEGIC ASSET

OCUP should not attempt to own every HSM, robot or accelerator. It should own the Runtime Authority architecture those systems are asked to enforce—and the evidence standard by which enforcement is judged.

9. Claims Discipline

Credibility is a strategic asset. All pilot materials should distinguish demonstrated evidence from future production capability.

Use	Avoid
Supports defensible incident reconstruction.	Shields the company from liability.
Demonstrates deterministic behavior under tested conditions.	Proves the system is unhackable or perfectly safe.
Creates a pathway toward hardware-backed enforcement.	Provides unalterable enforcement today.
May support future certification, underwriting and standards work.	Is already industry-standard or pre-certified.
Pre-production evidence and technical due diligence.	Production deployment or safety certification.

10. Invitation to Become an Early Runtime Authority Partner

The first objective is not a broad deployment. It is a bounded, paid engagement that answers one high-value question for the enterprise:

THE PILOT QUESTION

Can OCUP convert our most consequential autonomous-system authority risks into deterministic, reviewable evidence—and reveal a credible path to production enforcement?

An early partner receives a company-specific evidence report, benchmark results, validator and lease records, integration-readiness findings and a strategic production roadmap. Where

warranted, the engagement may lead to licensing, joint development, standards collaboration, strategic investment or M&A discussions.

Preserving humanity’s seat at the table begins with something concrete: proving that machines cannot grant themselves the right to continue beyond the boundaries humans establish.

Selected Internal Technical Sources

- OCUP Architecture Overview: temporal leases, validator consensus, append-only audit, unified validator service and production HSM requirements.
- OCUP Demo Scenarios: self-extension denial, lease expiry, human-authorized renewal, bridge quarantine, wallet protection and unified validator operation.
- OCUP Unified Dashboard CLI: shared validator pool, audit events and multi-domain scenario execution.
- OCUP Runtime Authority Evidence Pilot materials: local deterministic evidence scope, paid-pilot positioning and production-boundary disclosures.

STANDALONE COMMERCIAL LEAVE-BEHIND

OCUP RUNTIME AUTHORITY EVIDENCE PILOTS

Pre-production benchmark, audit and technical due-diligence engagements for autonomous systems.

Mission: Preserve human authority by proving that autonomous systems cannot indefinitely extend, enlarge or restore their own operational authority.

THE OPERATING LINE

pilot.ocup.ai runs the challenge. evidence.ocup.ai proves what happened.

Program	Price / Duration	Best For	Core Deliverables
Tier 1 Reference Evidence Pilot	\$35K–\$45K 90 days	Insurers, strategic evaluators, emerging robotics companies and fleet operators.	Standard benchmark suite; company-specific Evidence Report; lease, denial, expiry and validator records; benchmark scorecard; executive closeout.
Tier 2 Integration Evidence Pilot	\$75K–\$125K 90–120 days	Robotics OEMs, autonomous fleets, infrastructure operators and technical underwriting teams.	Everything in Tier 1; client-specific scenarios; telemetry mapping; adversarial definitions; integration-readiness assessment; production-path recommendations.
Tier 3 Strategic Anchor Program	\$150K–\$250K+ 120–180 days	Major OEMs, insurers, defense contractors, cloud platforms and ecosystem partners.	Everything in Tiers 1 and 2; custom benchmark family; fleet scenarios; HSM/enforcement mapping; distributed-validator roadmap; licensing and strategic integration plan.

Signature Deliverable

SELF-EXTENSION DENIAL PROOF

A deterministic evidence record showing that an autonomous system attempted to continue or enlarge its authority beyond an authorized temporal boundary—and that the request was denied without relying on the system’s voluntary compliance.

Every program is explicitly pre-production. The engagement does not constitute live production safety infrastructure, insurance approval or third-party certification.

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